

CCT College Dublin Continuous Assessment

Programme Title:	<i>BSc in Computing in IT</i>		
Delivery Mode:	FT		
Cohort Details:	<i>BSc (Hons) in Computing in IT - Sep 2022 cohort, Semester 8</i>		
Module Title(s):	<i>Problem Solving for Industry</i> <i>Programme schedules are all published on the CCT IQR Provider Profile</i>		
Assignment Type:	<i>Individual/ Pair</i>	Weighting(s):	90%
Assignment Title:	<i>CA2 Project</i>		
Lecturer(s):	<i>Muhammad Iqbal, Ken Healy</i>		
Issue Date:	<i>24th Feb 2026</i>		
Submission Deadline Date:	<i>29th May 2026 (23:59 Irish Standard Time)</i>		
Late Submission Penalty:	Late submissions will be accepted up to 5 calendar days after the deadline. All late submissions are subject to a penalty of 10% <u>of the mark awarded</u>. Submissions received more than 5 calendar days after the deadline above <u>will not</u> be accepted and a mark of 0% will be awarded.		
Method of Submission:	This assignment is submitted via Moodle. • <i>Project Report</i> • <i>Poster Presentation</i> • <i>Solution/Artefact/Findings Demonstration</i>		
Instructions for Submission:	Upload separate files on Moodle including MS word report, Jupyter notebook, R/ Python code, Poster presentation, Presentation, and any other supporting documents. <i>No pdf and zip files will be accepted.</i>		
Feedback Method:	Results posted in Moodle gradebook		
Feedback Date:	<i>Results will be announced after the exams board.</i>		

Assessment Outline

GitHub Link (90%):

<https://classroom.github.com/a/DZqk1zVt>

- **GitHub Classroom Requirement:** All students must use GitHub Classroom for version control. Python/R code must be developed and documented in a Jupyter Notebook or R file, with clear evidence of ongoing progress and individual contribution. The findings, analysis, and discussion must be compiled in an MS Word report.
- **Moodle Submission Requirement:** All required assignment files must be uploaded to Moodle before the deadline. Files stored only on GitHub will not be accepted as a valid Continuous Assessment (CA) submission.

Assessment Task

The capstone project will be a group-based endeavor where learners will explore a data analytics or artificial intelligence (AI) domain of their choice. Using the CRISP-DM (Cross Industry Standard Process for Data Mining) framework, students will identify and address a real-world problem within these fields. The pairs will be formed within the first two to three weeks of the programme based on a formation technique based on the cohort size, for example, randomisation, prior knowledge and experience or self-selection. The group size will be based on pair (Max 2 group members). Group members will collaborate using a GitHub classroom account with the module facilitators added as additional group members for monitoring and feedback purposes. Group members can use additional collaboration tools that they believe will enhance their pair communication and interaction, but GitHub classroom remains the only recognised repository for pair work for assessed purposes. The link for Github classroom is mentioned below

Note: The students are allowed to work alone or independently based on the permission by the module lecturer.

The initial area to be investigated should be an area of interest to all within each group. Facilitators will engage in these initial ideas and help groups scope the project to manageable proportions. This process ensures learners are working towards deadlines and facilitates guidance as new learning and knowledge is acquired as the curriculum progresses. The structure of the project in phases is expressed in the below:

Deliverable	Description	Learner Activity
Deliverable 1	Project Report	Learner groups are formed, and groups begin reflecting on the overarching area of interest, investigate data available, and initially express their understanding and concerns. Learner groups engage in pair formation and the separation of roles and responsibilities to address the six phases of the CRISP-DM framework, which are: 1. Business Understanding 2. Data Understanding 3. Data Preparation 4. Modelling 5. Evaluation 6. Deployment In addition, you should have an Introduction to the report with a section dedicated to Roles and Responsibilities. The report should also contain a Conclusion chapter with an Appendix containing a Group Reflection and an

		Appendix capturing evidence of group work.
Deliverable 2	Poster Presentation	Learner groups will design and develop a Poster Presentation to capture their approach to the six phases of the CRISP-DM framework.
Deliverable 3	Solution/Findings Demonstration	Learner groups will present their working solution (Data visualisations, results, ML or AI models/ implementation, conclusions etc) in the form of a pre-recorded presentation. Learners will also be required to attend a live Q & A session where faculty members will raise specific questions relating to the documentation and solution produced.

Please review the CCT Grade Descriptor available in this CA Project for a detailed description of the standard of work required for each grade band.

The grading system in CCT is the QQI percentage grading system and is in common use in higher education institutions in Ireland. The pass mark and thresholds for different grade bands may be different from what you have experience of in the higher education system in other countries. CCT grades must be considered in the context of the grading system in Irish higher education and not assumed to represent the same standard the percentage grade reflects when awarded in an international context.

Summary of Assessed Deliverables and Submission Dates

Deliverables	Assessed Element	Description	% Weighting	
			Individual Contribution	Group Contribution
Deliverable 1: Project Report (Due 29 th May 2026 @11.59pm)	CRISP-DM Implementation	Learner groups will develop a project plan with roles and responsibilities to address their area of interest. The group will develop their plan following the CRISP-DM framework and associated phases.	10%	5%
	Group Reflection Report	The group will be required to complete an end of project summary reflection report.	10%	n/a
	Deliverable 1 Total Marks (25%)		20%	5%
Deliverable 2: Poster Presentation (Due 29 th May 2026 @11.59pm)	Design, Content, Layout, Articulation	Learner groups will create a Poster Presentation to succulently capture their phased approach to addressing the overarching project problem and proposed solution.	20%	5%
	Deliverable 2 Total Marks (25%)		20%	5%

Deliverable 3: Solution/Findings Demonstration (Due 29th May 2026 @11.59pm. LIVE Q&A Session to be held on 8th June and 9th 2026 between 10am and 4.00pm at CCT College)	Working Solution	Learner groups will present their working solution (Data visualisations, results, AI models/ implementation, conclusions etc.) in the form of a pre-recorded presentation.	10%	10%
	Q&A Session	All group members will be asked to attend a live Q&A session where questions will be raised by faculty members.	20%	n/a
	Deliverable 3 Total Marks (40%)		30%	10%
TOTAL MARKS			70%	20%

Assessment Requirements

All assessment submissions must meet the following minimum requirements:

- Formative feedback will be provided in-class as the project progresses, and group monitoring will be conducted through the designated GitHub classroom account accounts.
- Your documentation should present your approach to the project, including elements of project planning (timelines).
- Ensure that your documentation follows a logical sequence through the planning/ research/ justification/ implementation phases of the project.
- Integrated/ Separate Jupyter notebooks/ Code files can be provided to show the code developed for the capstone project.
- Please ensure that additional resources are placed and linked to a logical file structure e.g., Scripts, Images, Report, Data etc...
- Ensure that you include your raw and structured datasets in your submission.
- 5000 (+/- 10%) words in report (not including code, code comments, titles, references or citations)
- Your Word count MUST be included in the report at the start.
- Use [Harvard Referencing](#) when citing third party material.
- Be the student's own work.
- Include the CCT assessment cover page.
- Read carefully the Artificial Intelligence policy (GenAI) policy for this integrated assignment.

Learning Outcomes:

Please note this is not the assessment task. The task to be completed is detailed on the next page.

This CA will assess student attainment of the following minimum intended learning outcomes:

- **MLO1:** Demonstrate an advanced ability to research an assigned problem area and propose an innovative solution while applying, professional, ethical and legal considerations. (Linked to PLO1)
- **MLO2:** Construct a detailed project report using appropriate scheduling techniques and perform a resource and risk analysis using various best practice approaches, with consideration for costs and benefits of both design and implementation choices. (Linked to PLO2)
- **MLO3:** Apply knowledge and understanding within specific areas in modern computing such as programming, databases, networking, web development and mobile technologies, in both familiar and unfamiliar domains. (Linked to PLO3)

- **MLO4:** Design and build a working prototype with consideration for various systems analysis and systems development approaches and formulate judgements and conclusions on design and performance with regard to possible trade-offs inherent in a given problem. (Linked PLO4)
- **MLO5:** Act under supervision to reflect on the project progression through the iterative design process and through the various supervised feedback mechanisms, and through peer feedback within teams. (Linked to PLO6)
- **MLO6:** Exhibit competence, accountability and autonomy by making informed critical decisions at key project milestones and perform relevant research required for in-depth understanding and awareness of project attributes to manage learning tasks independently. (Linked to PLO7)
- **MLO8:** Confidently present a working product in a succinct, efficient manner and show how the product was developed with specific reference to sectorial relevance, identify areas for improvement and further development. (Linked to PLO8)

Use of Artificial Intelligence

Artificial Intelligence (AI) can be used in this assignment in the following way(s):

- Brainstorming and refining your ideas;
- Fine tuning your research questions;
- Finding information on your topic;
- Drafting an outline to organise your thoughts;
- Checking grammar and style
- Other

Note for students:

- You are required to include all AI prompt(s) and outputs in an appendix, including a short reflection (approx. 250 words from each student, 500 words from pair) on how AI enhanced your work, or otherwise. These words are not part of 5000 words asked for CA.
- You are responsible for the information you submit based on an AI query. Your use of AI tools must be properly documented and cited.
- Any assignment that is found to have used generative AI tools in an unauthorised way will be subject to college disciplinary procedures as outlined in the QA Manual.
- Cite all AI references used in your assignment.
- You must declare how you used AI on your assignment cover sheet.
- In the case of any suspicious work submitted by a student, the student will be asked questions related to GenAI usage in the Q and A session.
- When in doubt about permitted usage, please ask for clarification to your supervisor.

Rubric

Marking Criteria	Individual %	Group %	Excellent (+70%)	Very Good (60–69%)	Good (50–59%)	Acceptable (40–49%)	Fail (<39%)
Deliverable 1: Project Report – CRISP-DM Implementation & Project Planning	10%	5%	Comprehensive and clearly structured application of CRISP-DM phases; clear objectives; well-defined roles; strong justification of approach and decisions; clear documentation of Business Understanding, Data Understanding, Data Preparation, Modelling,	Clear and logically structured application of all CRISP-DM phases; objectives defined; roles explained; sound justification provided though critical depth or integration between phases may be limited.	Most CRISP-DM phases are addressed with reasonable structure; objectives stated but may lack precision; roles described but limited analytical justification of decisions.	Basic or uneven coverage of CRISP-DM phases; objectives vague; limited linkage between phases; roles minimally explained; justification largely descriptive.	Poor or missing CRISP-DM understanding; phases missing/incorrect; no roles and no justification provided.

Marking Criteria	Individual %	Group %	Excellent (+70%)	Very Good (60–69%)	Good (50–59%)	Acceptable (40–49%)	Fail (<39%)
			Evaluation and Deployment.				
Deliverable 1: Group Reflection Report	10%	n/a	Deep critical reflection on teamwork, challenges, learning outcomes and improvements; clear evidence of collaboration (e.g., GitHub contributions); strong evaluation of individual and pair performance.	Clear reflection on teamwork and learning; identifies challenges and improvements; some evaluative commentary but limited critical depth.	Reflection describes teamwork and outcomes; limited critical evaluation; focus mainly on activities rather than learning impact.	Minimal reflection; largely descriptive summary of activities; little evidence of critical thinking or learning evaluation.	Missing or superficial reflection; no clear evidence of learning or collaboration.
Deliverable 2: Poster Presentation – Design, Content, Layout & Articulation	20%	5%	Highly professional poster; clear structure; strong visuals and data visualisations; succinct communication of all CRISP-DM phases; logical narrative from problem to solution; visually engaging and academically presented.	Well-structured poster with clear communication of key CRISP-DM phases; visuals appropriate and supportive; minor clarity or formatting limitations.	Poster communicates main elements of the project; structure generally clear; visuals present but may lack refinement or strong narrative coherence.	Basic poster with limited visual hierarchy; explanation of phases lacks clarity; visuals minimal or weakly integrated into the narrative.	Poor design and clarity; missing or irrelevant visuals; fails to communicate approach or solution effectively.
Deliverable 3: Working Solution / Findings Demonstration (Pre-recorded Presentation)	10%	10%	Fully working solution; clear data visualisations and results; strong ML/AI justification; independent and dependent variables clearly identified; appropriate feature selection; hyperparameter tuning evident; clear train/test split; strong evaluation metrics; k-fold cross-validation applied; meaningful model comparisons and critical interpretation of findings.	Functional solution with clear explanation of methodology and results; variables identified; appropriate validation techniques applied; interpretation demonstrates understanding though critical comparison may lack depth.	Working solution with adequate explanation; basic validation applied; metrics presented; limited discussion of model selection, tuning, or comparative performance.	Partially functional solution; methodology described at a surface level; limited or inconsistent validation; metrics poorly interpreted.	Poor or missing ML understanding and justification; variables not identified. Poor feature selection; no tuning; missing or irrelevant metrics. Poor or missing train/test split and cross-validation. No meaningful comparison, visualisations, or statistical understanding.
Deliverable 3: Live Q&A Session	20%	n/a	All members demonstrate excellent understanding; confident, technically accurate, and reflective responses; able to justify modelling and design decisions; demonstrate ownership of individual contribution.	Demonstrates strong understanding of the project and methodology; responses mostly accurate and justified; minor gaps in depth or clarity.	Demonstrates general understanding; able to answer core questions; technical explanations may lack depth or precision.	Limited understanding evident; responses partially correct but lack clarity, confidence, or technical accuracy.	Poor understanding; unable to answer effectively; minimal participation or unclear ownership of work.
Deliverable 3 Total	30%	10%					
OVERALL TOTAL	70%	20%					

Reassessment (General Policy)

Different but pedagogically equivalent and appropriately weighted repeat assessment tasks may be given for repeat assessments. Learners who fail to achieve the minimum intended learning outcomes associated with a given assessment will receive constructive feedback (in addition to the normal assignment feedback) on the submitted work and the brief of the repeat assessment. In the case of group work, the group may be directed to recover the failed element of a group assignment, and/or individuals within groups may be directed to recover the individually assessment elements of a group project.

The Irish Grading System

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Additional Information

- Lecturers are not required to review draft assessment submissions. This may be offered at the lecturer's discretion.
- In accordance with CCT policy, feedback to learners may be provided in written, audio or video format and can be provided as individual learner feedback, small group feedback or whole class feedback.
- Results and feedback will only be issued when assessments have been marked and moderated / reviewed by a second examiner.
- Additional feedback may be provided as individual, small group or whole class feedback. Lecturers are not obliged to respond to email requests for additional feedback where this is not the specified process or to respond to further requests for feedback following the additional feedback.
- Following receipt of feedback, where a student believes there has been an error in the marks or feedback received, they should avail of the recheck and review process and should not attempt to get a revised mark / feedback by directly approaching the lecturer. Lecturers are not authorised to amend published marks outside of the recheck and review process or the Board of Examiners process.
- Students are advised that disagreement with an academic judgement is not grounds for review.
- For additional support with academic writing and referencing students are advised to contact the CCT Library Service.
- For additional support with subject matter content students are advised to contact the [CCT Student Mentoring Academy](#)
- For additional support with IT subject content, students are advised to access the [CCT Support Hub](#).